

Vectorization and Cython

1 Executing a Python Function

- a motivating example
- timing the execution

2 Speeding Up Python

- vectorization
- Cython

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Introduction to Symbolic Computation
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Approximating π

One way to approximate π is
to compute the area of the unit circle via numerical integration.

$$\frac{\pi}{4} = \int_0^1 \sqrt{1-x^2} \approx \frac{1}{n} \sum_{k=1}^n \sqrt{1 - \left(\frac{k}{n}\right)^2}$$

This way is certainly not the best way to approximate π .

But the formulas are simple and computationally intensive,
which serves as a good motivating example to speedup code.

The n is the parameter to control the cost of this computation.

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Timing the Execution

We distinguish between small and large computations.

- For small computations, in the order of milliseconds, the timings should execute a loop of several runs.
- For large computations:
 - 1 mark the start of the computations,
 - 2 do the computation, and
 - 3 compute the elapsed CPU time.

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Vectorization

Lists are versatile data structures, but arrays (vectors)

- 1 allow for compact storage, as each item is of the same type,
- 2 can be accessed with fast index arithmetic.

Vectorization replaces arithmetical operations on vectors with functions that take vectors as arguments.

Applied to $\sum_{k=1}^n \sqrt{1 - \left(\frac{k}{n}\right)^2}$ we do the following:

- 1 Make a vector x of equidistant points in $[0, 1]$.
- 2 Apply the function $\sqrt{}$ to the entire vector x .
- 3 Apply the function `sum` to $\sqrt{x}$.

The functions on vectors are *optimized* and *compiled*.

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Cython

Cython is a variant of Python:

- 1 It adds type declarations.
- 2 Calls functions directly from the C standard library.
- 3 Is compiled.

Compared to vectorization,

- cythonizing code leaves the original control structure intact,
- while vectorizing often requires a significant reformulation.